

Brandon Oubre, Ph.D.

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Research Interests

Mobile health and health informatics, with a focus on digital, quantitative behavioral assessment of neurologic disease signs.

Professional Experience

University of Alabama at Birmingham Department of Computer Science

Assistant Professor

Birmingham, Alabama

Aug. 2024 – Present

Harvard Medical School, Massachusetts General Hospital

Research Fellow in Neurology (PI: Dr. Anoopum S. Gupta)

Boston, Massachusetts

Sept. 2022 – Aug. 2024

UMass Amherst Manning College of Information and Computer Sciences

Ph.D. Student (Advised by Prof. Sunghoon Ivan Lee)

Amherst, Massachusetts

Sept. 2017 – Aug. 2022

Biogen Digital Health

Research Intern: Movement Data Analyst

Cambridge, Massachusetts

June 2021 – Aug. 2021

CenturyLink

Software Developer II

Monroe, Louisiana

June 2015 – Aug. 2017

NASA Johnson Space Center

Software Engineering Intern

Houston, Texas

May 2014 – July 2014

NASA Ames Research Center

Software Engineering Intern

Mountain View, California

June 2013 – Aug. 2013

Education

University of Massachusetts Amherst

PhD Computer Science **Outstanding Dissertation Award**

Amherst, Massachusetts

September 2022

University of Massachusetts Amherst

MS Computer Science

Amherst, Massachusetts

May 2020

Louisiana State University

BS Computer Science and BS Mathematics

Baton Rouge, Louisiana

May 2015

Honors and Awards

- 2024 **Outstanding Service Award**, IEEE EMBS Body Sensor Networks (BSN)
- 2024 **Featured Article**, IEEE Journal of Biomedical and Health Informatics
- 2022 **Outstanding Dissertation Award**, Manning College of Information and Computer Sciences
- 2022 **Featured Article**, IEEE Transactions on Biomedical Engineering
- 2022 **Dissertation Writing Fellowship**, Manning College of Information and Computer Sciences
- 2020 **Featured Article**, IEEE Transactions on Neural Systems and Rehabilitation Engineering
- 2019 **NSF GRFP Honorable Mention**, NSF Graduate Research Fellowship Program
- 2019 **NSF Student Registration Award**, IEEE EMBS BHI/BSN
- 2017 **Graduate School Fellowship**, Manning College of Information and Computer Sciences
- 2017 **James Kurose Scholar**, Manning College of Information and Computer Sciences
- 2015 **Outstanding Thesis Award**, Louisiana State University Honors College
- 2015 **University Medalist**, Louisiana State University
- 2014 **Barry M. Goldwater Scholar**, Barry Goldwater Scholarship and Excellence in Education Foundation
- 2014 **Official State Commendation**, Louisiana Senate Resolution SR39
- 2013 **Clayton Engineering Excellence Award**, Louisiana State University College of Engineering
- 2011 **LA-STEM Research Scholarship**, Louisiana State University Office of Strategic Initiatives

Journal Publications

- [J1] A. Nunes, S. Patel, **B. Oubre**, *et al.*, “Multimodal digital phenotyping of behavior in a neurology clinic: Development of the neurobooth platform and the first two years of data collection,” *medRxiv*, Dec. 2024. DOI: <https://doi.org/10.1101/2024.12.28.24319527>.
- [J2] J. Lee, **B. Oubre**, J.-F. Daneault, C. D. Stephen, J. D. Schmahmann, A. S. Gupta, and S. I. Lee, “Contrastive learning model for wearable-based ataxia assessment,” *IEEE Trans. Biomed. Eng.*, Sep. 2025. DOI: <https://doi.org/10.1109/TBME.2025.3608674>.
- [J3] **B. Oubre**, F. Yang, A. Luddy, R. Manohar, N. N. Soja, C. D. Stephen, J. D. Schmahmann, D. Kulkarni, L. White, S. Patel, and A. S. Gupta, “A digital measure of eye movements during reading sensitively captures oculomotor and speech dysfunction, early changes, and disease progression in ataxias,” *Ann. Neurology*, Sep. 2025. DOI: <https://doi.org/10.1002/ana.78039>.
- [J4] S. Patel, **B. Oubre**, C. D. Stephen, J. D. Schmahmann, and A. S. Gupta, “Submovements derived from wearable sensors capture ataxia severity and differ across motor tasks and directions of motion,” *Cerebellum*, vol. 24, no. 6, p. 156, Sep. 2025.
- [J5] K. Vattis, **B. Oubre**, A. C. Luddy, J. S. Ouillon, N. M. Eklund, C. D. Stephen, J. D. Schmahmann, A. S. Nunes, and A. S. Gupta, “Sensitive quantification of cerebellar speech abnormalities using deep learning models,” *IEEE Access*, vol. 12, pp. 62 328–62 340, Apr. 2024.
- [J6] **B. Oubre** and S. I. Lee, “Detection and assessment of point-to-point movements during functional activities using deep learning and kinematic analyses of the stroke-affected wrist,” *IEEE J. Biomed. Health Inform.*, vol. 28, no. 2, pp. 1022–1030, Feb. 2024, **[Featured Article]**.
- [J7] J.-F. Daneault, **B. Oubre**, J. G. V. Miranda, and S. I. Lee, “Understanding voluntary human movement variability through data-driven segmentation and clustering,” *Frontiers Human Neurosci.*, vol. 17, Nov. 2023.
- [J8] J. Lee, **B. Oubre**, J.-F. Daneault, S. I. Lee, and A. S. Gupta, “Estimation of ataxia severity in children with ataxia-telangiectasia using ankle-worn sensors,” *J. Neurology*, Jul. 2023.
- [J9] Y. Liu, **B. Oubre**, C. Duval, S. I. Lee, and J.-F. Daneault, “A kinematic data-driven approach to differentiate involuntary choreic movements in individuals with neurological conditions,” *IEEE Trans. Biomed. Eng.*, vol. 69, no. 12, pp. 3784–3791, Dec. 2022.
- [J10] J. Lee, **B. Oubre**, J.-F. Daneault, C. D. Stephen, J. D. Schmahmann, A. S. Gupta, and S. I. Lee, “Analysis of gait sub-movements to estimate ataxia severity using ankle inertial data,” *IEEE Trans. Biomed. Eng.*, vol. 69, no. 7, pp. 2314–2323, Jul. 2022.
- [J11] **B. Oubre**, S. Lane, S. Holmes, K. Boyer, and S. I. Lee, “Estimating ground reaction force and center of pressure using low-cost wearable devices,” *IEEE Trans. Biomed. Eng.*, vol. 69, no. 4, pp. 1461–1468, Apr. 2022, **[Featured Article]**.
- [J12] **B. Oubre**, J.-F. Daneault, K. Whritenour, N. C. Khan, C. D. Stephen, J. D. Schmahmann, S. I. Lee, and A. S. Gupta, “Decomposition of reaching movements enables detection and measurement of ataxia,” *Cerebellum*, vol. 20, no. 6, pp. 811–822, Dec. 2021.
- [J13] **B. Oubre**, J.-F. Daneault, K. Boyer, J. H. Kim, M. Jasim, P. Bonato, and S. I. Lee, “A simple low-cost wearable sensor for long-term ambulatory monitoring of knee joint kinematics,” *IEEE Trans. Biomed. Eng.*, vol. 67, no. 12, pp. 3483–3490, Dec. 2020.
- [J14] **B. Oubre**, J.-F. Daneault, H.-T. Jung, K. Whritenour, J. G. V. Miranda, J. Park, T. Ryu, Y. Kim, and S. I. Lee, “Estimating upper-limb impairment level in stroke survivors using wearable inertial sensors and a minimally-burdensome motor task,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 3, pp. 601–611, Mar. 2020, **[Featured Article]**.
- [J15] P. Khaloo, **B. Oubre**, J. Yang, T. Rahman, and S. I. Lee, “Nose: A novel odor sensing engine for ambient monitoring of the frying cooking method in kitchen environments,” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 3, no. 2, 49:1–49:25, Jun. 2019, ISSN: 2474-9567.

Conference Proceedings

- [C1] Y. Liu, **B. Oubre**, C. Duval, S. I. Lee, and J.-F. Daneault, “Submovement characteristics are related to the presence of parkinson’s disease,” in *47th Annu. Int. Conf. IEEE Eng. Medicine Biol. Soc.*, IEEE, Jul. 2025.
- [C2] **B. Oubre** and S. I. Lee, “Estimating post-stroke upper-limb impairment from four activities of daily living using a single wrist-worn inertial sensor,” in *IEEE EMBS Int. Conf. Biomed. Health Inform.*, IEEE, Sep. 2022.

- [C3] **B. Oubre**, J.-F. Daneault, H.-T. Jung, J. Park, T. Ryu, Y. Kim, and S. I. Lee, “Estimating quality of reaching movement using a wrist-worn inertial sensor,” in *42nd Annu. Int. Conf. IEEE Eng. Medicine Biol. Soc.*, IEEE, Jul. 2020.

Abstracts, Talks, and Posters

- [A1] M. Z. Chowdury, S. Patel, D. Kulkarni, C. D. Stephen, J. D. Schmahmann, A. S. Gupta, and **B. Oubre**, “Sequence-aware models of eye movements during passage reading enables more precise assessment of ataxia,” in *22nd IEEE Int. Conf. Wearable Implantable Body Sensor Netw.*, Nov. 2025.
- [A2] **B. Oubre**, F. Yang, A. Luddy, R. Manohar, N. N. Soja, C. D. Stephen, J. D. Schmahmann, D. Kulkarni, L. White, S. Patel, and A. S. Gupta, “In-clinic eye tracking during passage reading supports precise assessment of oculomotor signs of ataxia,” in *Int. Congress for Ataxia Res.*, **[Flash Talk]**, Nov. 2024.
- [A3] **B. Oubre**, F. Yang, A. Luddy, R. Manohar, N. N. Soja, C. D. Stephen, J. D. Schmahmann, D. Kulkarni, L. White, S. Patel, and A. S. Gupta, “In-clinic eye tracking during passage reading supports precise assessment of oculomotor signs of ataxia,” in *21st IEEE Int. Conf. Wearable Implantable Body Sensor Netw.*, Oct. 2024.
- [A4] **B. Oubre**, “Digital and quantitative behavioral phenotyping in neurologic disease,” *ML4Health Seminar Series, Broad Institute*, Feb. 2024, **[Invited Talk]**.
- [A5] **B. Oubre**, J.-F. Daneault, K. Whritenour, N. C. Khan, C. D. Stephen, J. D. Schmahmann, S. I. Lee, and A. S. Gupta, “Decomposition of reaching movements enables detection and measurement of ataxia,” *2nd Annu. Massachusetts General Hospital Ataxia Center Symp.*, May 2021, **[Invited Talk]**.
- [A6] **B. Oubre**, K. Whritenour, J.-F. Daneault, A. S. Gupta, and S. I. Lee, “Estimation of ataxia severity using wrist-worn sensors and the finger-to-nose test,” *Nat. Ataxia Found. Ataxia Investigators Meeting*, Mar. 2020.
- [A7] **B. Oubre**, K. Whritenour, J.-F. Daneault, A. S. Gupta, and S. I. Lee, “Estimation of ataxia severity using wrist-worn sensors and the finger-to-nose test,” *1st Annu. Massachusetts General Hospital Ataxia Center Symp.*, Mar. 2020, **[Invited Talk]**.
- [A8] J. Yang, A. Varga, K. Tung, A. Chandra, **B. Oubre**, N. Ramasarma, E. K. Choe, P. Bonato, and S. I. Lee, “A finger-worn sensor network for monitoring the real-world performance of stroke survivors,” *16th IEEE Int. Conf. Wearable Implantable Body Sensor Netw.*, May 2019.

Academic Service

Conference Organizing

- 2025 **Student Activities Chair**, IEEE EMBS Body Sensor Networks (BSN)
- 2025 **Clinical Abstracts Chair**, IEEE EMBS Body Sensor Networks (BSN)
- 2023–2024 **Publicity Chair**, IEEE EMBS Body Sensor Networks (BSN)

Editorial

- 2023–2025 **Associate Editor**, IEEE Int. Engineering in Medicine and Biology Conf. (EMBC)

Peer Review

IEEE JBHI, IEEE TNSRE, JNER, Footwear Sci., Sci. Reports, NPJ Parkinson’s Disease, Cerebellum, Sensors, Frontiers Bioeng. Biotechnol.

Teaching

CS 685/785: Foundations of Data Science

Instructor

UAB

Spring 2025

CS 467: Machine Learning Fundamentals

Instructor

UAB

Fall 2025

CS 350: Automata and Formal Languages

Instructor

UAB

Fall 2024–2025

COMPSCI 590W / INFO 390W: Health Informatics and Data Science

Teaching Assistant

UMass Amherst

Spring 2021–2022

COMPSCI 240: Reasoning Under Uncertainty

Lead Teaching Assistant

UMass Amherst

Fall 2021

Teaching Assistant

Fall 2020